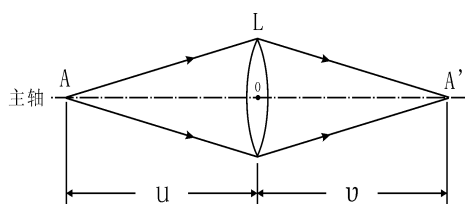


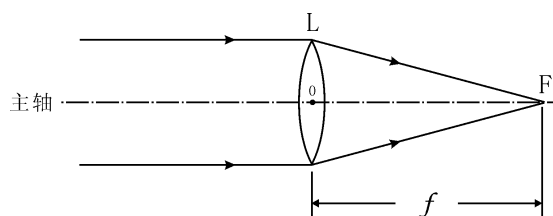
Measurement of focal lengths of thin lenses

INTRODUCTION

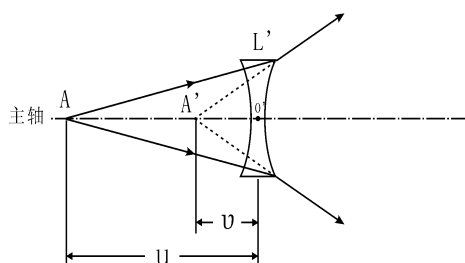
Lenses are of two basic types, convex (or positive) which are thicker in the middle than at the edges and concave(or negative) for which the reverse holds. The passage of light through a lens can be described by following the paths of a few selected rays. A lens or system of lenses have an axis of symmetry called the optical (or principal) axis. Rays that head toward each other are called converging; rays that head away from each other are called diverging. Parallel rays of light passing through a converging or positive lens converge at a point (called the focal point) in the focal plane, forming a real image(Fig.1(a)). Parallel rays passing through a concave or negative lens diverge and form a virtual image (Fig.1(b)). The position of the image can be calculated using the equation $1/u + 1/v = 1/f$ (where u = the distance from the object to the lens, v = the distance from the image to the lens, and f = the focal length of the lens)



(a)

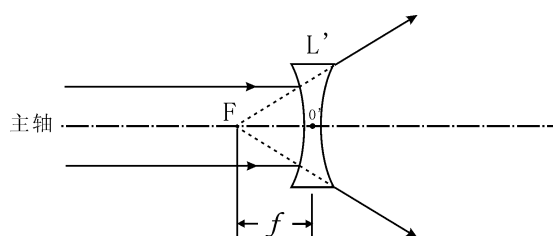


(a)



(b)

Fig.(1)



(b)

Fig.(2)

For the present our treatment will be confined to paraxial rays, i.e. rays close to the axis and making very small angles with it. In effect this means we shall only consider lenses of small aperture but in diagrams both angles and lenses will be made larger for clarity.

The principal focus F of a thin lens is the point on the principal axis, towards which all paraxial rays ,parallel to the principal axis, converge in the case of a convex lens or from which they appear to diverge in the case of a concave lens, after refraction, Fig.2. Since light can fall on either surface , a lens has two principal foci, one on each side, and these are equidistant from centre O. The distance FO is the focal length of the lens. A convex lens is a converging lens and has real foci. A concave lens is a diverging lens and has virtual foci.

The focal length is important for thin lenses .We will measure the focal length of a convex lens using different methods.

EXPERIMENT: measuring f of thin lenses

1. Converging lens

(a) Plane mirror method.

Using the arrangement shown in Fig.3, a pin or illuminated object is adjusted until it coincide in position with its image, located no parallax or by a screen. The rays from the object must emerge from the lens and fall on the plane mirror normally to retrace their path. The object is therefore at the principal focus.

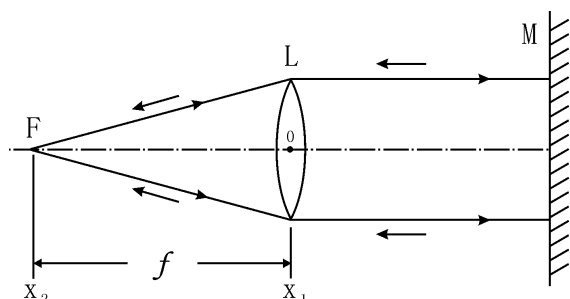


Fig.3

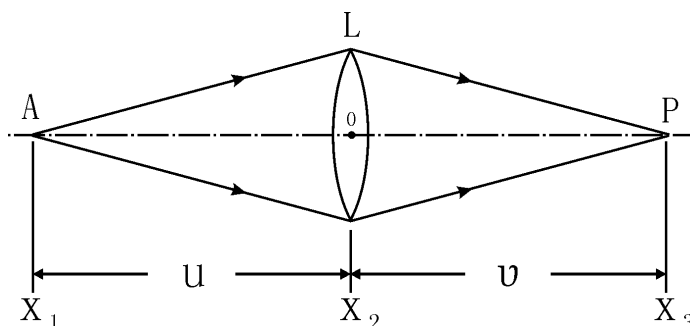


Fig.4

(b) lens formula method

Several values of the image distance v , corresponding to different values of the object distance u are found using either pin/no parallax method or an illuminated object and screen, Fig.4. For each pair of values, f is calculated from $1/f = 1/v + 1/u$ and the average taken.

A better plan is to plot a graph of $1/v$ against $1/u$, draw the best straight line through the points. The intercepts on the axes are both equal to $1/f$ since when $1/u = 0, 1/v = 1/f$ or when $1/v = 0, 1/u = 1/f$, from the simple lens equation.

Complete table 1.

(c) Two-position method.

The image I of an illuminated object A is obtained on a screen P , Fig.5. With the object and screen in the same position, the lens is moved from I to II where another sharp image is obtained. (The image is magnified for position I of the lens and diminished for II , why?) The distances between A and P , D , and between I and II , l , are measured then f calculated from

$$f = \frac{D^2 - l^2}{4D}$$

By changing D , a set of values of D and l can be found to give an average value of f . Complete table 2.

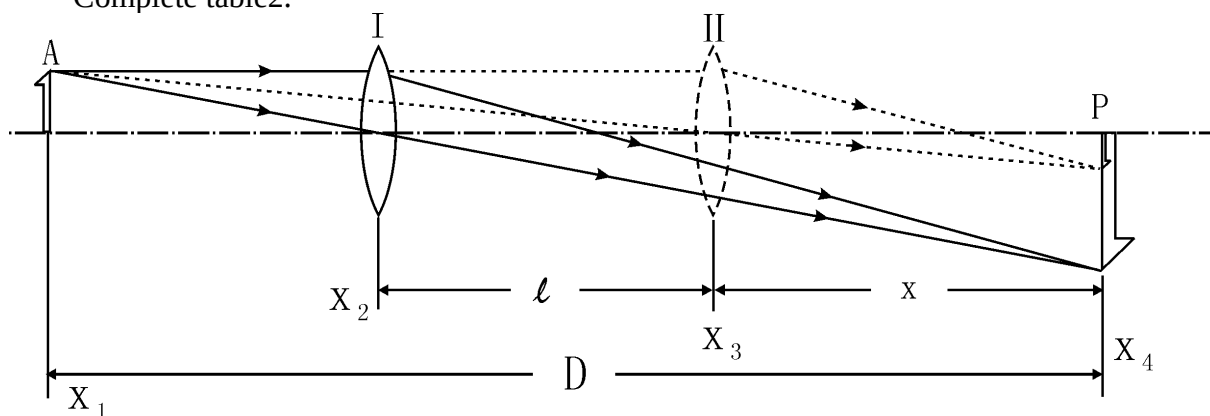


Fig.5

Note. A converging lens cannot form a real image on a screen if (i) the object is inside the

principal focus or (ii) the distance between the object and the screen is less than $4f$ (this can be proved theoretically and experimentally). When the separation is $4f$, the object and image are then each distant $2f$ from the lens on opposite sides. In the 'two-position' method for f , the separation of the object and screen must exceed $4f$.

2. Diverging lens

Auxiliary converging lens method.

A diverging lens normally forms a virtual image of a real object. Such an image cannot be located by a screen and is not easily found by pin/no parallax method. However with the help of a converging lens, a real image can be obtained.

In Fig.6 the converging lens forms a real image at P_1 of an object A when the diverging lens is absent. This image is located and its position noted. The diverging lens is then placed between L_1 and P_1 and the converging beam of light falling on it behaves as a virtual object at P_1 . The diverging lens forms a real image of P_1 at P_2 , which is located. For the diverging lens $u = L_2 P_1$ and $v = L_2 P_2$ from which f can be calculated.

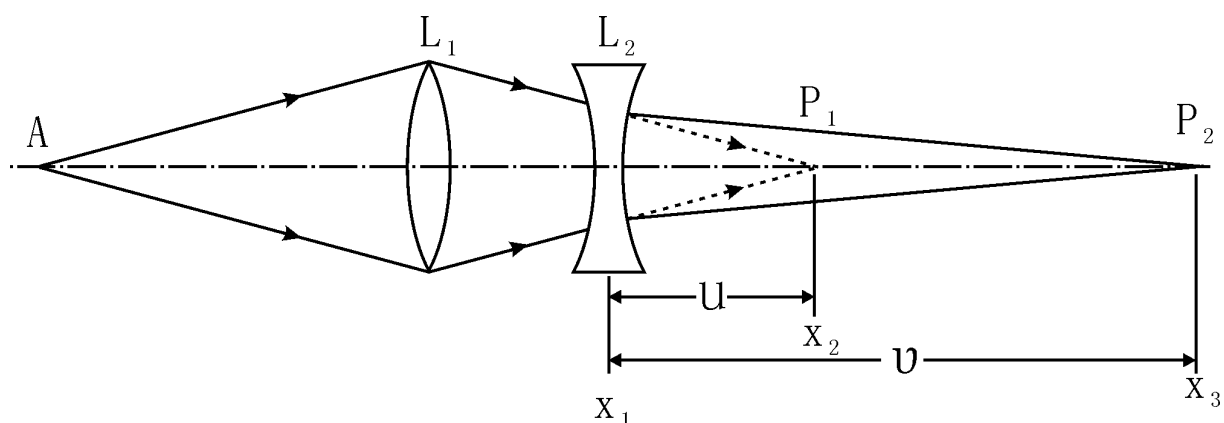


Fig.6

By changing L_2 , a set of values of u and v can be found to give an average value of f .
Complete table3

DATA

1. Converging lens

(a) Plane mirror method.

$f =$ _____

(b) lens formula method

table1

	x_1	x_2	x_3	u	v	f
1						
2						
3						
4						
5						
平均						

Plot the graph $1/v$ against $1/u$

From the graph $f =$ _____

(c) Two-position method.

Table2

	x_1	x_2	x_3	x_4	D	l	f
1							
2							
3							
4							
5							
平均							

2. Diverging lens

Auxiliary converging lens method.

Table3

	x_1	x_2	x_3	u	v	f
1						
2						
3						
平均						

PRE-QUESTION

Do you know other methods to measure the focal length of converging or diverging lens?
Please write out your idea.

QUESTION

1. What should you make prepare before you measuring the focal length?
2. Which method is more accurate? why